

When More Isn't Better: The Role of Relationship Quality in Cognitive Aging

2021127027 Chae Yeon Kim

Abstract

This study examined how the quantity and quality of social relationships - This study examined how both the quality and quantity of close social relationships - including support and burden from spouses, family members/relatives, and friends—are associated with cognitive function among older adults in Korea. Using data from the first wave of the Korean Social Life, Health, and Aging Project (KSHAP), MMSE scores were analyzed among 814 participants. Higher spousal and friend support were linked to better cognitive function, whereas greater relational burden from spouses and family members was associated with poorer outcomes. While the number of friends showed a marginal positive association with cognition, family size alone was unrelated; instead, the cognitive benefit of family ties depended on their emotional quality, as larger networks were linked to poorer outcomes when family support was low. Furthermore, friend support moderated the negative effect of depressive symptoms on cognition. These findings underscore the importance of fostering supportive, low-strain relationships across all domains to promote cognitive resilience in later life.

Introduction

Cognitive decline poses a significant public health challenge in aging societies. While pharmacological treatments remain limited, recent research highlights the vital role of emotionally meaningful social relationships in preserving cognitive health. Both positive aspects of close ties—such as emotional support—and negative aspects—such as relational burden—have been linked to cognitive outcomes. Moreover, depressive symptoms are known to contribute to cognitive decline, with social support potentially buffering these effects.

Prior research has demonstrated that both the structural and functional characteristics of social relationships contribute to cognitive aging. Holtzman et al. (2004) found that larger and more active social networks provide cognitive stimulation and are linked to slower cognitive decline. Beyond network size, Seeman et al. (2001) emphasized that relationship quality—particularly emotional support—plays a more influential role in preserving cognitive function, while relational strain can have detrimental effects.

Furthermore, a recent study by de Sousa et al. (2024) showed that the source of social support—whether from spouse, children, relatives, or friends—differentially affects mental health outcomes in older adults, highlighting the importance of distinguishing between relationship types. Recent work by Roh et al. (2022) further demonstrated that social support can buffer the negative effects of depressive symptoms on cognitive outcomes.

However, few studies have systematically compared how both the quality (support vs. burden) and quantity of different close relationships interact to shape cognitive outcomes. Moreover, little is known about whether the protective effects of supportive relationships vary across relationship types, or how relational burden may moderate these associations—particularly in East Asian contexts where family-centered norms may introduce unique relational dynamics.

To address these gaps, this study uses data from the first wave of the Korean Social Life, Health, and Aging Project (KSHAP) to examine how social relationships are associated with cognitive function in later life. Specifically, this study addresses the following research questions:

(1) How are the *quality* of close relationships—measured by emotional support and burden from spouses, family/relatives, and friends—associated with cognitive function?

(2) How is the *quantity* of social ties (number of friends, number of family/relatives) associated with cognitive function?

(3) Does support from friends buffer the association between depressive symptoms and cognitive decline?

Data

This study uses data from the first wave of the Korean Social Life, Health, and Aging Project (KSHAP). The survey was conducted in 2011 and targeted all individuals aged 60 and over, along with their spouses (regardless of age), residing in a village (the smallest administrative unit) in Township K, Ganghwa Island, South Korea. A total of 814 respondents participated in the first wave.

Measurements

Covariates - In all regression models, age, sex, education level, income, and marital status, smoking, Instrumental Activities of Daily Living were controlled.

IADL impairment was coded as “Impairment” for participants who responded “Extremely”, “Slightly”, or “Often” on the item (“During the past 4 weeks, how often did your physical symptoms disturb your daily works”). For the question “Do you smoke cigarettes?”, participants who responded “Yes”, “Recently yes, but no in the past” were regarded as smokers.

Depressive symptoms - Depressive symptoms were assessed using 20 items from the Korean CES-D scale, with three positively worded items reverse coded and all items rescaled from 1–4 to 0–3. The total score ranged from 0 to 60, with higher scores indicating more severe symptoms.

Cognitive Measure - Cognitive change was computed by using total MMSE scores (0–30). "Not measured" was treated as "Incorrect," whereas in Wave 3, "Not measured" was treated as "Don't know." Responses marked as "Not measured" were treated as incorrect responses and included as such in the analysis. Following Kim et al.(2001), who reported an optimal MMSE-K cut-off score of 21/22 for dementia screening in Korean older adults, participants were categorized as having either Normal Cognition (MMSE $\geq$ 21) or Suspected Cognitive Impairment (MMSE < 21) in this study. The 25th percentile MMSE score in our sample was 21.

Social Support - Social support variables were constructed from survey items on spouse, family/relatives, and friends/neighbors. Friend support, family support, and spouse support scores were calculated as the mean of two items (communication and reliance for help) within each relationship domain. Likewise, friend burden, family burden, and spouse burden scores were calculated as the mean of two items (unreasonable demands and blame) within each domain. All items were rated on a 4-point Likert scale (1 = Never, 2 = Rarely, 3 = Sometimes, 4 = Often), with "Don't know" responses coded as missing. An overall emotional support score was calculated as the mean of friend, family, and spouse support scores; an overall emotional burden score was calculated as the mean of friend, family, and spouse burden scores.

Quantitative aspects of social relationships

- (1) Sum of social ties, calculated as the sum of close family members/relatives, friends, and spouse (if present), using the midpoint of each response category;
- (2) Number of close family members/relatives, and
- (3) Number of close friends.

Response options for the number of close family members/relatives and friends were categorized as follows: 0 (None), 1 (1 person), 2.5 (2–3 persons), 6.5 (4–9 persons), 15 (10–20 persons), and 21 (more than 21 persons). "Don't know" responses were treated as missing.

Family support, family burden, and number of family members/relatives refer to perceived interactions with family members and relatives (excluding spouse).

Analysis

Table 1. Characteristics of Study Participants by MMSE Group (with Overall)				
Characteristic	Overall N = 814 ¹	Normal Cognition N = 654 ¹	Suspected Cognitive Impairment N = 160 ¹	p-value ²
MMSE (Cognitive Function)	23.62 ± 5.34	25.61 ± 2.73	15.46 ± 5.64	<0.001
CES-D (Depressive Symptoms)	11.51 ± 8.26	10.46 ± 7.33	15.80 ± 10.27	<0.001
Sum of Social Ties	10.41 ± 7.04	10.66 ± 7.18	9.38 ± 6.33	0.031
No. of Family & Relatives	6.40 ± 5.39	6.37 ± 5.29	6.54 ± 5.84	0.6
No. of Friends	3.24 ± 3.30	3.47 ± 3.41	2.31 ± 2.60	<0.001
Emotional Support	2.87 ± 0.57	2.89 ± 0.56	2.78 ± 0.59	0.053
Emotional Burden	1.74 ± 0.59	1.75 ± 0.60	1.71 ± 0.57	0.6
Spouse Support	3.47 ± 0.64	3.50 ± 0.62	3.25 ± 0.77	0.003
Spouse Burden	1.98 ± 0.76	1.98 ± 0.76	2.01 ± 0.71	0.5
Friend Support	2.45 ± 0.76	2.48 ± 0.75	2.34 ± 0.81	0.11
Friend Burden	1.63 ± 0.66	1.63 ± 0.66	1.63 ± 0.65	0.8
Family/Relative Support	2.87 ± 0.77	2.85 ± 0.77	2.95 ± 0.78	0.2
Family/Relative Support	1.72 ± 0.68	1.73 ± 0.69	1.71 ± 0.62	0.7
Gender				0.001
Male	342 (42%)	295 (45%)	47 (29%)	
Female	472 (58%)	359 (55%)	113 (71%)	
Education				<0.001
No formal education	242 (30%)	137 (21%)	105 (66%)	
Elementary	334 (42%)	287 (44%)	47 (30%)	
Middle School	113 (14%)	108 (17%)	5 (3.1%)	
High School	84 (10%)	83 (13%)	1 (0.6%)	
College or higher	31 (3.9%)	30 (4.7%)	1 (0.6%)	
Income				0.016
Low	412 (63%)	333 (61%)	79 (75%)	
Mid-low	153 (24%)	133 (24%)	20 (19%)	
Mid	62 (9.5%)	55 (10%)	7 (6.6%)	
High	24 (3.7%)	24 (4.4%)	0 (0%)	
Smoking				0.061
Non-smoker	716 (88%)	568 (87%)	148 (93%)	
Smoker	98 (12%)	86 (13%)	12 (7.5%)	
IADL Impairment				<0.001
No Impairment	594 (73%)	513 (79%)	81 (51%)	
Impairment	216 (27%)	137 (21%)	79 (49%)	

¹ Mean ± SD; n (%)

² Wilcoxon rank sum test; Pearson's Chi-squared test with simulated p-value (based on 2000 replicates); Fisher's Exact Test for Count Data with simulated p-value (based on 2000 replicates)

Table 1 presents the descriptive statistics of all study participants (N = 814), as well as stratified comparisons between two groups : Normal Cognition and Suspected Cognitive Impairment.

Significant group differences were observed in MMSE scores, depressive symptoms (CES-D), sum of social ties, number of friends, emotional burden, spouse support, gender,

education, income, and IADL impairment (all $p < 0.05$). Emotional support was marginally significant. ($p=0.053$)

Individuals with suspected cognitive impairment had significantly lower MMSE scores and significantly higher depressive symptoms (CES-D), suggesting a close link between cognitive decline and mental health. They also had fewer total social ties and friends, highlighting a potential pattern of social withdrawal or network contraction among cognitively impaired individuals. Notably, spouse support was also significantly lower, possibly indicating weaker intimate support networks in this group. Gender, education, and income levels were also associated with cognitive status: cognitively impaired individuals were more likely to be female, less educated, and from lower-income brackets — consistent with known socioeconomic risk factors for cognitive decline. Additionally, IADL impairment was more prevalent in this group, further suggesting that functional limitations co-occur with cognitive impairment.

Variable	MMSE	CES-D	Emotional Support	Emotional Burden	Spouse Support	Spouse Burden	Friend Support	Friend Burden	Family Support	Family Burden	No. Friends	No.Family & Relatives
MMSE		-0.29 *	0.12 *	-0.1 *	0.14 *	-0.1 *	0.1 *	-0.07	0.01	-0.08	0.19 *	-0.04
CES-D	-0.29 *		-0.15 *	0.31 *	-0.13 *	0.31 *	-0.11 *	0.23 *	-0.09 *	0.23 *	-0.17 *	0
Emotional Support	0.12 *	-0.15 *		0.24 *	0.57 *	0.08 *	0.75 *	0.3 *	0.81 *	0.25 *	0.17 *	-0.01
Emotional Burden	-0.1 *	0.31 *	0.24 *		0.02	0.86 *	0.31 *	0.84 *	0.18 *	0.87 *	-0.03	-0.2 *
Spouse Support	0.14 *	-0.13 *	0.57 *	0.02		-0.04	0.08	0.04	0.22 *	0.05	0.14 *	0.14 *
Spouse Burden	-0.1 *	0.31 *	0.08 *	0.86 *	-0.04		0.12 *	0.55 *	0.08	0.62 *	-0.08	-0.17 *
Friend Support	0.1 *	-0.11 *	0.75 *	0.31 *	0.08	0.12 *		0.42 *	0.46 *	0.26 *	0.13 *	-0.1 *
Friend Burden	-0.07	0.23 *	0.3 *	0.84 *	0.04	0.55 *	0.42 *		0.17 *	0.62 *	0.03	-0.16 *
Family Support	0.01	-0.09 *	0.81 *	0.18 *	0.22 *	0.08	0.46 *	0.17 *		0.21 *	0.1 *	-0.05
Family Burden	-0.08	0.23 *	0.25 *	0.87 *	0.05	0.62 *	0.26 *	0.62 *	0.21 *		-0.01	-0.18 *
No. Friends	0.19 *	-0.17 *	0.17 *	-0.03	0.14 *	-0.08	0.13 *	0.03	0.1 *	-0.01		0.25 *
No.Family & Relatives	-0.04	0	-0.01	-0.2 *	0.14 *	-0.17 *	-0.1 *	-0.16 *	-0.05	-0.18 *	0.25 *	

Above table presents the bivariate correlations among cognitive function (MMSE), depressive symptoms (CES-D), network characteristics, and social support and burden variables.

Table 2 presents the bivariate correlations among cognitive function (MMSE), depressive symptoms (CES-D), network characteristics, and social support and burden variables. MMSE scores were positively associated with network size, emotional support, spouse support, friend support, and close friends and family/relatives, and negatively associated with emotional burden, spouse burden, and CES-D scores.

CES-D scores were negatively associated with network size, emotional support, spouse support, friend support, family support, and close friends and family/relatives, and positively associated with emotional burden, spouse burden, friend burden, and family burden.

Emotional support was also positively correlated with emotional burden, indicating that supportive relationships may still entail some relational strain.

These results suggest that both the quantity (network size, close friends, family/relatives) and quality (support and burden) of social relationships are closely linked to cognitive functioning and mental health in older adults.

Variable	Model 1 (Control only) Coef. (SE)	Model 1 (Control only) p-value	Model 2-1 (Spouse) Coef. (SE)	Model 2-1 (Spouse) p-value	Model 2-2 (Friend) Coef. (SE)	Model 2-2 (Friend) p-value	Model 2-3 (Family) Coef. (SE)	Model 2-3 (Family) p-value
Intercept	20.029*** (0.582)	0.000	20.296*** (1.188)	0.000	20.741*** (0.864)	0.000	22.743*** (0.947)	0.000
Has Spouse	1.871*** (0.455)	0.000			1.583*** (0.457)	0.001	1.777*** (0.450)	0.000
Gender: Female	0.544 (0.389)	0.163	0.547 (0.368)	0.137	0.484 (0.389)	0.215	0.417 (0.377)	0.269
Education: Elementary School	2.892*** (0.432)	0.000	2.280*** (0.454)	0.000	2.587*** (0.432)	0.000	2.890*** (0.420)	0.000
Education: Middle School	4.644*** (0.582)	0.000	4.229*** (0.575)	0.000	4.329*** (0.585)	0.000	4.387*** (0.566)	0.000
Education: High School	5.015*** (0.654)	0.000	4.441*** (0.634)	0.000	4.511*** (0.656)	0.000	4.684*** (0.635)	0.000
Education: College or Higher	4.990*** (0.939)	0.000	3.485*** (0.918)	0.000	3.884*** (0.980)	0.000	4.202*** (0.917)	0.000
Income: Mid-Low	0.142 (0.413)	0.732	0.858* (0.407)	0.035	0.124 (0.411)	0.763	-0.005 (0.401)	0.990
Income: Middle	1.143 (0.592)	0.054	1.807** (0.563)	0.001	0.825 (0.597)	0.167	0.871 (0.577)	0.132
Income: High	1.084 (0.932)	0.245	1.141 (0.867)	0.189	0.926 (0.924)	0.316	1.032 (0.895)	0.249
Smoker	-0.221 (0.518)	0.670	-0.133 (0.491)	0.786	-0.151 (0.513)	0.769	-0.231 (0.503)	0.647
IADL Impairment	-2.029*** (0.403)	0.000	-0.865* (0.436)	0.048	-1.460*** (0.422)	0.001	-1.220** (0.413)	0.003
Spousal Support			0.942*** (0.268)	0.000				
Spousal Burden			-0.542* (0.234)	0.021				
Friend Support					0.539* (0.246)	0.029		
Friend Burden					-0.531 (0.294)	0.071		
Family Support							-0.124 (0.220)	0.573
Family Burden							-0.512* (0.255)	0.045
Number of Friends					0.093 (0.052)	0.076		
Number of Family/Relatives							-0.031 (0.031)	0.314
Depressive Symptoms (CES-D)			-0.064* (0.025)	0.012	-0.093*** (0.023)	0.000	-0.099*** (0.022)	0.000

The dependent variable of all regression models was the Mini-Mental State Examination (MMSE) score, used as an indicator of cognitive function. Model 1 included only control variables, such as gender, marital status, education, income, smoking, and IADL impairment. Model 2 sequentially added relational variables: spousal support and burden (Model 2-1), friend support, burden, and the number of friends (Model 2-2), family and relative support, burden and the number of families and relatives (Model 2-3), and overall emotional support and burden (Model 2-4). All models except model 1 also included depressive symptoms (CES-D) to account for mental health effects.

As shown in Table 3, social relationships were significantly associated with cognitive function. In the control model (Model 1), being married ($\beta = 1.87$, $p < .001$), higher education, and absence of IADL impairment were positively associated with MMSE scores.

When spousal relationship variables were included (Model 2-1), both higher spousal support ($\beta = 0.94$, $p < .001$) and lower spousal burden ($\beta = -0.54$, $p = .021$) were significant predictors of better cognitive function.

Similarly, friend support in Model 2-2 ($\beta = 0.54$, $p = .029$) was positively associated with MMSE, while family burden in Model 2-3 was negatively associated ($\beta = -0.51$, $p = .045$), suggesting that relationship strain within family contexts may harm cognitive outcomes. Notably, across all models, higher depressive symptoms were strongly linked to poorer cognitive function (all $p < .001$). These findings underscore the importance of relationship quality—particularly the balance between support and burden—in shaping cognitive health.

Regarding the quantity of relationships, the number of friends showed a marginally positive association with cognitive function ($\beta = 0.093$, $p = .076$), indicating that the quantity of friendships may contribute to cognitive health. In contrast, the number of family/relatives was unrelated to cognitive outcomes. This may be because an increase in family size does not necessarily equate to greater emotional closeness or positive interactions; rather, it may involve additional obligations or caregiving burdens, which can negate potential cognitive benefits.

Table 4. Emotional Support and Burden on Cognitive Function

Variable	Model 2-4 (Emotional Support / Burden) Coef. (SE)	Model 2-4 (Emotional Support / Burden) p-value
Intercept	21.059*** (1.052)	0.000
Has Spouse	1.607*** (0.463)	0.001
Gender: Female	0.507 (0.383)	0.186
Education: Elementary School	2.689*** (0.429)	0.000
Education: Middle School	4.486*** (0.579)	0.000
Education: High School	4.685*** (0.647)	0.000
Education: College or Higher	4.076*** (0.937)	0.000
Income: Mid-Low	0.123 (0.407)	0.763
Income: Middle	0.826 (0.591)	0.163
Income: High	0.878 (0.917)	0.339
Smoker	-0.126 (0.511)	0.806
IADL Impairment	-1.381*** (0.417)	0.001
Emotional Support	0.547. (0.313)	0.081
Emotional Burden	-0.661* (0.309)	0.033
Depressive Symptoms (CES-D)	-0.101*** (0.023)	0.000

As shown in Table 4, when overall emotional support and burden were included (Model 2-4), emotional burden remained a significant negative predictor of cognitive function ($\beta = -0.66$, $p = .033$), while emotional support was positively associated but not statistically significant ($\beta = 0.55$, $p = .081$).

Consistent with previous models, being married, higher education, and absence of IADL impairment were positively associated with MMSE scores. In addition, higher depressive symptoms remained strongly associated with lower cognitive function ($\beta = -0.10$, $p < .001$).

These results suggest that emotional strain across relationships, rather than perceived support alone, plays a critical role in maintaining cognitive health in later life.

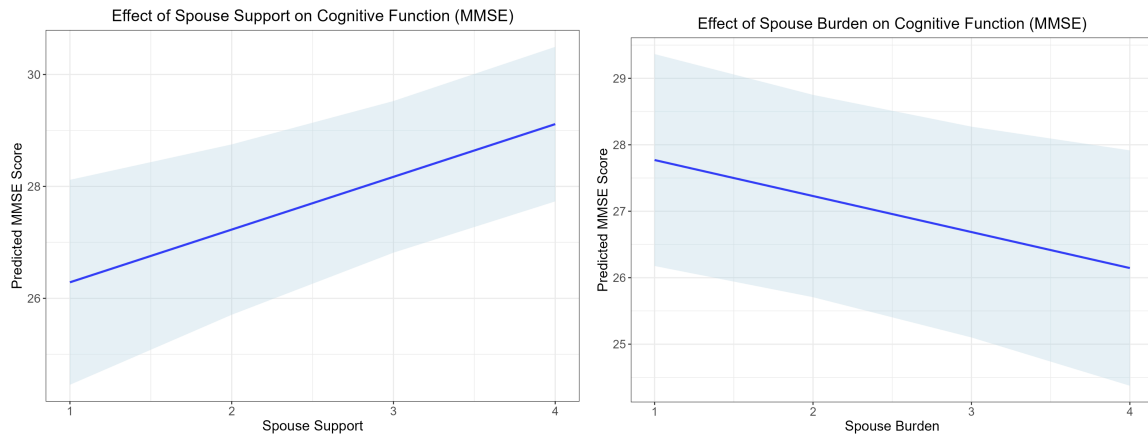
Variable	Model 3-1 (Family Num × Family Support) Coef. (SE)	Model 3-1 (Family Num × Family Support) p-value	Model 3-2 (Friend Support × CES-D) Coef. (SE)	Model 3-2 (Friend Support × CES-D) p-value
Intercept	22.100*** (0.739)	0.000	20.327*** (0.586)	0.000
Has Spouse	1.773*** (0.449)	0.000	1.816*** (0.456)	0.000
Gender: Female	0.429 (0.377)	0.255	0.504 (0.393)	0.201
Education: Elementary School	2.875*** (0.420)	0.000	2.657*** (0.436)	0.000
Education: Middle School	4.354*** (0.566)	0.000	4.266*** (0.589)	0.000
Education: High School	4.667*** (0.634)	0.000	4.600*** (0.660)	0.000
Education: College or Higher	4.213*** (0.916)	0.000	4.728*** (0.967)	0.000
Income: Mid-Low	-0.054 (0.401)	0.894	0.172 (0.414)	0.679
Income: Middle	0.850 (0.576)	0.141	1.205* (0.595)	0.043
Income: High	0.990 (0.894)	0.269	1.322 (0.929)	0.155
Smoker	-0.178 (0.503)	0.724	-0.070 (0.519)	0.893
IADL Impairment	-1.255** (0.413)	0.002	-2.006*** (0.407)	0.000
Family Burden	-0.440 (0.258)	0.088		
Number of Family/Relatives (centered)	-0.020 (0.032)	0.537		
Family Support (centered)	-0.110 (0.220)	0.617		
Family Num × Family Support	0.063 (0.037)	0.089		
Friend Support (centered)			0.409 (0.222)	0.066
Friend Support × CES-D			0.091*** (0.024)	0.000
Depressive Symptoms (CES-D)	-0.098*** (0.022)	0.000		

As shown in Table 5, both interaction models partially supported differential effects of relationship characteristics on cognitive function.

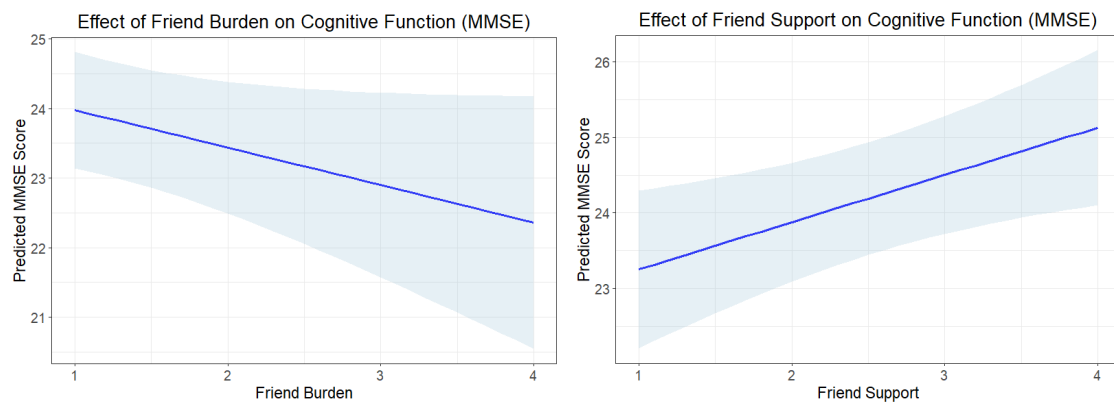
In Model 3-1, the interaction term was marginally significant ($\beta = 0.063$, $p = .089$), suggesting that the effect of family support on cognitive function may depend on the size of one's family network. Specifically, while family support was only modestly associated with MMSE in smaller networks, its beneficial effects became more pronounced as the number of family members increased. Conversely, in cases where family support was low, having a larger family network was associated with poorer cognitive outcomes, possibly due to relational strain or unmet expectations within family relationships. This pattern highlights the contingent nature of family ties, where both their quantity and quality jointly influence cognitive function in later life.

In contrast, Model 3-2 showed a significant positive interaction ($\beta = 0.091$, $p < .001$), indicating that the protective effect of friend support on cognitive function was more pronounced in individuals with higher depressive symptoms. This pattern suggests that friend support may serve as a protective factor that buffers the negative impact of depression on cognitive function.

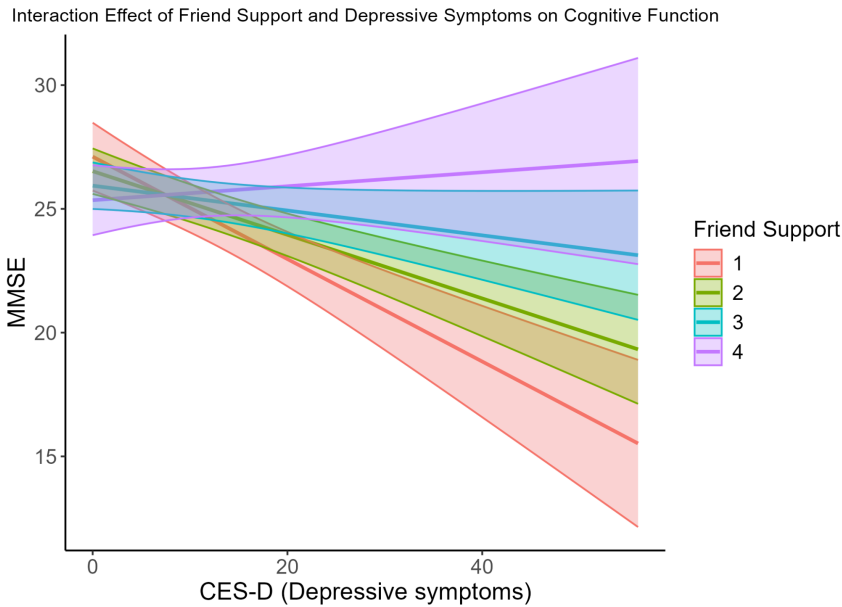
In both models, depressive symptoms (CES-D) consistently showed a negative association with MMSE scores ($p < .001$), reaffirming the strong link between mental health and cognitive outcomes. Additionally, functional impairment (IADL) remained a robust negative predictor of cognitive function across models. Physical health, measured as IADL, was controlled in the analyses because impairments in daily functioning are known to be closely linked to cognitive decline in older adults. Such impairments may restrict opportunities for social interaction and cognitive stimulation, while also contributing to depressive symptoms and overall reductions in cognitive reserve.



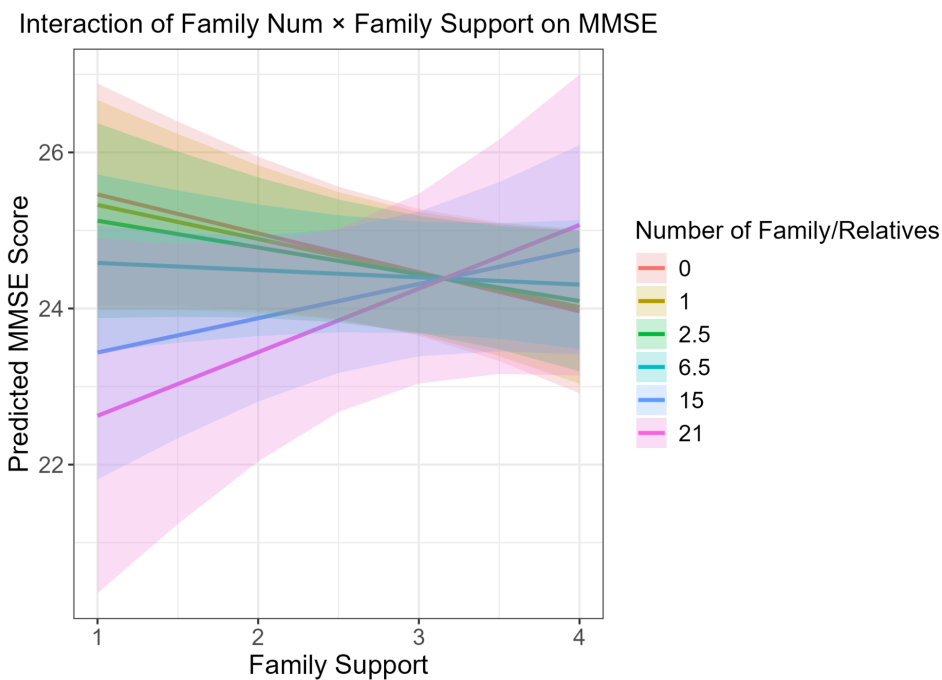
Higher levels of spousal support were associated with better cognitive function, as indicated by higher predicted MMSE scores (left panel). In contrast, greater spousal burden was linked to lower cognitive function (right panel). This suggests that positive partner interactions promote cognitive health, whereas relational strain may impair it.



Above tables display the main effects of friend burden and friend support on cognitive function (MMSE). Higher friend burden was associated with lower MMSE scores, whereas greater friend support was positively associated with cognitive function. These results suggest that the quality of friendships has a significant impact on cognitive health.



Above figure presents the interaction effect between friend support and depressive symptoms (CES-D) on cognitive function (MMSE). At low levels of friend support (pink line), higher depressive symptoms were strongly associated with lower cognitive function. In contrast, at high levels of friend support (purple line), this negative association between depressive symptoms and MMSE was substantially buffered. These findings indicate that *greater perceived friend support may mitigate the adverse cognitive effects of depression*, highlighting the protective role of supportive friendships for cognitive health in older adults.



Above figure illustrates the interaction between family support and the number of family/relatives on cognitive function (MMSE). The interaction between family network size and family support revealed that the protective effect of family support on cognitive function was contingent upon the number of family members.

When family support was low, a larger number of family members was associated with lower cognitive function, suggesting that having more family members can become a source of burden when emotional support is lacking. In contrast, when family support was high, a larger family network was linked to better cognitive function, indicating that having more supportive family members may serve as a protective factor for cognitive health.

Discussion & Conclusions

This study addressed how the quality and quantity of close relationships shape cognitive function in later life, and whether friend support buffers the cognitive risks of depression.

First, relationship quality—measured by support and burden—was a consistent and significant predictor across all relationship types. Higher spousal and friend support were associated with better cognitive function, whereas greater relational burden from spouses and family members predicted poorer outcomes.

Second, the quantity of social ties showed mixed effects. A greater number of friends was marginally linked to better cognitive outcomes, but a larger family network was not directly beneficial; rather, the cognitive effects of family ties depended on emotional quality. When family support was low, having more family members was associated with poorer cognitive outcomes, suggesting that, without adequate support, larger family networks may place additional emotional or caregiving demands on older adults.

Third, friend support buffered the negative impact of depressive symptoms on cognitive function, indicating that friendships may play a protective role in mitigating the cognitive risks associated with depression.

Overall, these findings highlight the importance of fostering emotionally supportive, low-strain relationships—whether with spouses, family members, or friends—as a key strategy for promoting cognitive resilience in older adults. Given the limited efficacy of pharmacological treatments for cognitive decline, public health efforts should prioritize programs that help older adults build and maintain high-quality relationships as part of broader strategies for healthy cognitive aging.

<Reference>

Kim JM, Shin IS, Yoon JS, Kim JH, Lee HY. Cut-off Score on MMSE-K for Screening of Dementia in Community Dwelling Old People. *J Korean Geriatr Psychiatry*. 2001;5(2):163-168.

Holtzman, R. E., Rebok, G. W., Saczynski, J. S., Kouzis, A. C., Wilcox Doyle, K., & Eaton, W. W. (2004). Social network characteristics and cognition in middle-aged and older adults. *The Journals of Gerontology Series B: Psychological Sciences and Social Sciences*, 59(6), P278–P284. <https://doi.org/10.1093/geronb/59.6.p278>

Seeman, T. E., Lusignolo, T. M., Albert, M., & Berkman, L. (2001). Social relationships, social support, and patterns of cognitive aging in healthy, high-functioning older adults: MacArthur studies of successful aging. *Health Psychology*, 20(4), 243–255. <https://doi.org/10.1037/0278-6133.20.4.243>

Rabelo, D. F., & Pinto, J. M. (2024). Social support network, functional capacity and mental health in older adults. *Psico-USF*, 28, 767-781.

Roh, H. W., Cho, E. J., Son, S. J., & Hong, C. H. (2022). The moderating effect of cognitive function on the association between social support and depressive symptoms among community-dwelling older adults: Cross-sectional and longitudinal analyses. *Journal of Affective Disorders*, 318, 185-190.